

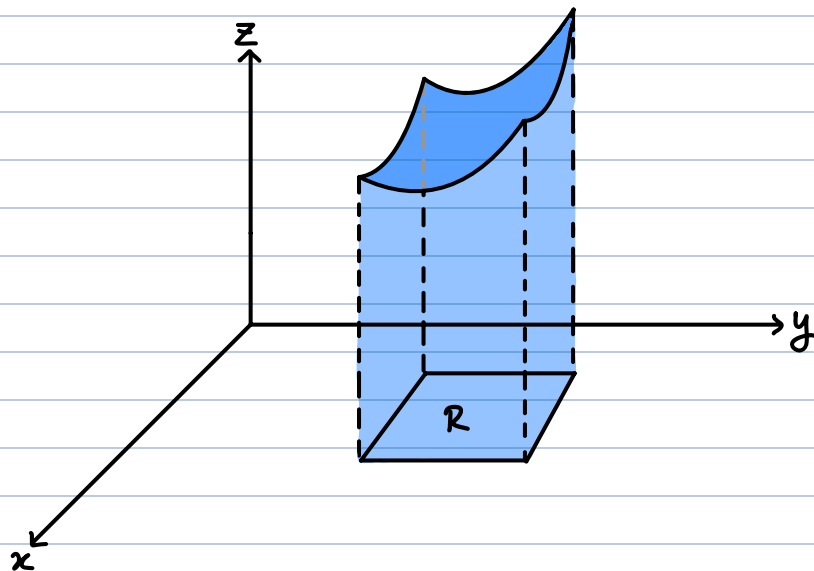
Double Integral:

Double integral of a function over the rectangle gives the volume below the surface $z = f(x, y)$ and above the rectangle R .

$$V = \iint_R f(x, y) \, dA$$

$$\Rightarrow V = \iint_R f(x, y) \, dx \, dy$$

$$\Rightarrow V = \iint_R f(x, y) \, dy \, dx$$



These are called iterated integrals

(e.g.) Evaluate the iterated integrals:

a) $\int_0^3 \int_1^2 x^2 y \, dy \, dx$

b) $\int_1^2 \int_0^3 x^2 y \, dx \, dy$

Solⁿ a) $\int_0^3 \int_1^2 x^2 y \, dy \, dx$

$$= \int_0^3 \left[\frac{x^2 y^2}{2} \right]_1^2 dx$$

$$= \int_0^3 \left[\frac{x^2 \cdot 4}{2} - \frac{x^2 \cdot 1}{2} \right] dx$$

$$= \int_0^3 2x^2 - \frac{1}{2}x^2 \, dx = \int_0^3 \frac{3x^2}{2} \, dx = \left[\frac{x^3}{2} \right]_0^3 = \frac{3^3}{2} - \frac{0^3}{2} = \frac{27}{2}$$

← Since dy comes first, the inner limits $\int_1^2$ are limits of y and the outer limits $\int_0^3$ are limits of x

$$b) \int_1^2 \int_0^3 x^2 y \, dx \, dy$$

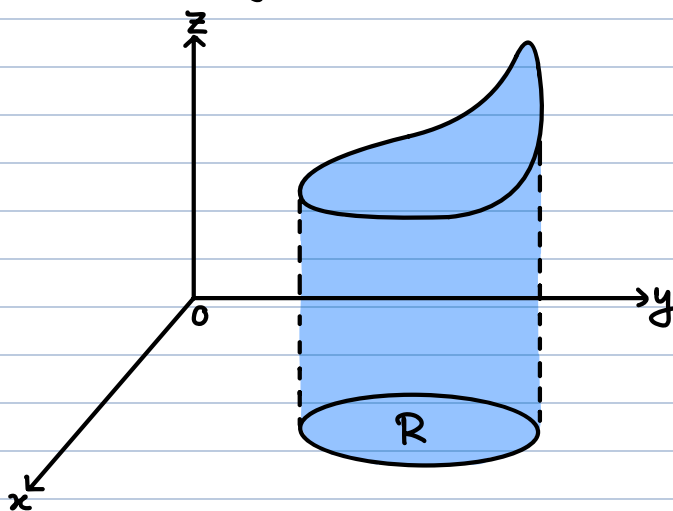
$$= \int_1^2 \left[\frac{x^3 y}{3} \right]_0^3 dy = \int_1^2 9y \, dy = \left[\frac{9y^2}{2} \right]_1^2 = \frac{36}{2} - \frac{9}{2}$$

$$= \frac{27}{2} //$$

Double integrals over general regions:

There are times where the region, R , over which you need to integrate is not a rectangle with sides parallel to the axes.

e.g. When the region R is a circle:



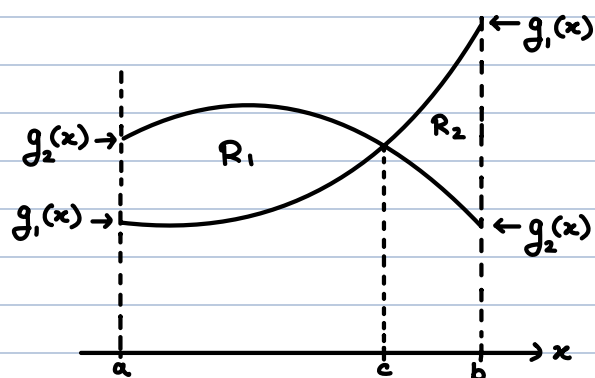
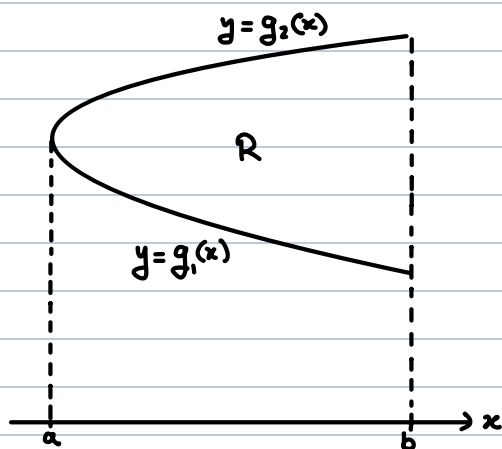
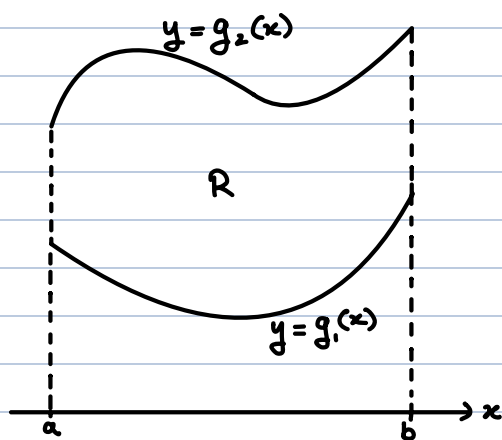
In this case, the limits of the integral are also functions.

* Note that the outer limits are still only scalars.

↳ Regions here can be divided into two types:

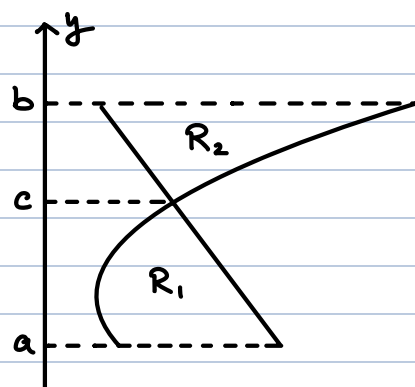
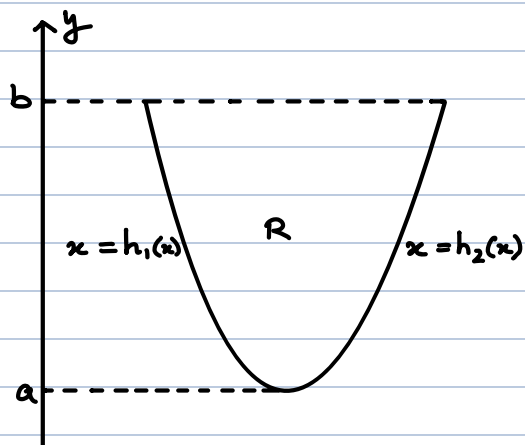
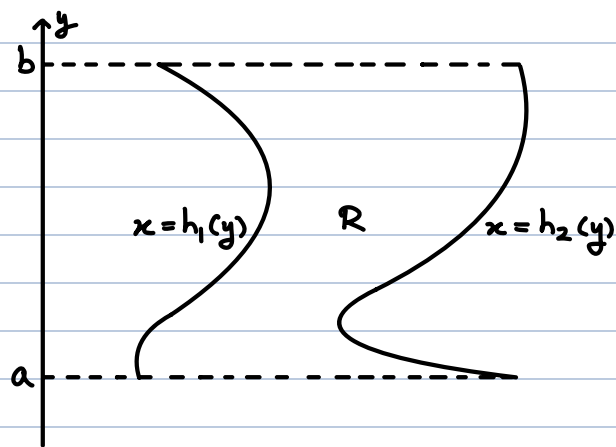
→ Type I Regions:

Regions bounded on the left and right sides by lines $x=a$ and $x=b$.



→ Type II Region:

Regions bounded on the top and bottom sides by lines $y=a$ and $y=b$.



* Notice the last two diagrams: there being an intersection between $g_1(x)$ and $g_2(x)$ results in two regions, R_1 and R_2 for type I. Similarly, due to an intersection between $h_1(x)$ and $h_2(x)$, there are also two regions for type II.

(e.g.) Evaluate $\iint_R xy \, dA$ over the region R , enclosed between $y = \frac{x}{2}$, $y = \sqrt{x}$, $x = 2$ and $x = 4$.

Soln Here, $z = f(x, y) = xy$

We first need to identify the type of region R is.

Since the limits of y are functions and the limits of x are constants, this is a type I region.

→ For the type I function, we then need to check if the limit functions of y intersect in the interval $(2, 4)$ in x :

$$\frac{x}{2} = \sqrt{x} \quad \Rightarrow \quad x = \frac{x^2}{4} \quad \Rightarrow \quad x^2 - 4x = 0$$

$$\Rightarrow x(x-4) = 0 \quad \Rightarrow \quad x = 0 \quad | \quad \Rightarrow \quad x = 4$$

None of these points lie inside the interval $(2, 4)$

in x . So we need not worry about an intersection.

→ Since there are no points in $(2, 4)$ where

$\frac{x}{2} = \sqrt{x}$, one must be greater than the other.

Taking a point inside $(2, 4)$, let $x=3$.

∴ At $x=3$, $y = \frac{(3)}{2} = \frac{3}{2}$ and $y = \sqrt{3}$.

As $\sqrt{x} > \frac{x}{2}$, we can say that $y = \frac{x}{2}$ is the lower limit, while $\frac{x}{2}$ is the upper limit.

→ For type I, $V = \iint_R f(x, y) dy dx$

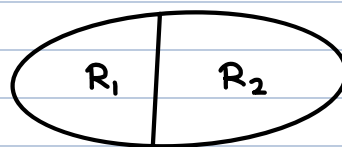
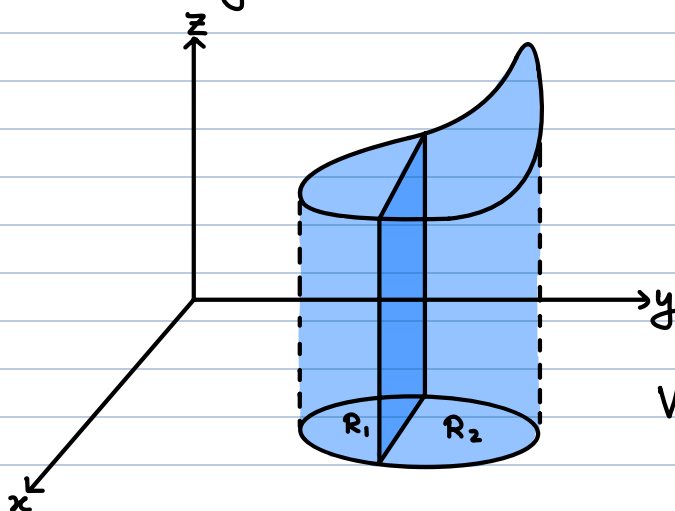
$$= \int_2^4 \int_{x/2}^{\sqrt{x}} xy dy dx = \int_2^4 \left[\frac{xy^2}{2} \right]_{x/2}^{\sqrt{x}} dx$$

$$= \int_2^4 \frac{x^2}{2} - \frac{x^3}{8} dx = \left[\frac{x^3}{6} - \frac{x^4}{32} \right]_2^4$$

$$= \left(\frac{4^3}{6} - \frac{4^4}{32} \right) - \left(\frac{2^3}{6} - \frac{2^4}{32} \right)$$

$$= \frac{8}{3} - \frac{5}{6} = \frac{11}{6}$$

Integration can at times be split into two:



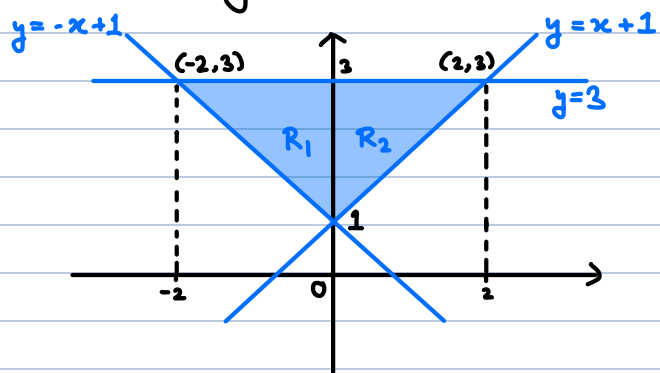
Then, volume is given by:

$$V = \iint_R f(x, y) dA = \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA$$

(e.g.) Evaluate: $\iint_R 2x - y^2 \, dA$ over the triangular region R

enclosed between the lines $y = x+1$, $y = -x+1$ and $y=3$.

Solⁿ: Drawing the region R :



← The region can be divided into two parts

$$\text{Then: } V = \iint_R 2x - y^2 \, dA$$

$$= \iint_{R_1} 2x - y^2 \, dA + \iint_{R_2} 2x - y^2 \, dA$$

$$= \int_{-2}^0 \int_{-x+1}^3 2x - y^2 \, dy \, dx + \int_0^2 \int_{x+1}^3 2x - y^2 \, dy \, dx$$

$$= \int_{-2}^0 \left[2xy - \frac{y^3}{3} \right]_{-x+1}^3 dx + \int_0^2 \left[2xy - \frac{y^3}{3} \right]_{x+1}^3 dx$$

$$= \int_{-2}^0 (6x - 9) - \left(-2x^2 + 2x - \frac{(-x+1)^3}{3} \right) dx$$

$$+ \int_0^2 (6x - 9) - \left(2x^2 + 2x - \frac{(x+1)^3}{3} \right) dx$$

$$= \int_{-2}^0 6x - 9 + 2x^2 - 2x + \frac{(-x+1)^3}{3} dx$$

$$+ \int_0^2 6x - 9 - 2x^2 - 2x + \frac{(x+1)^3}{3} dx$$

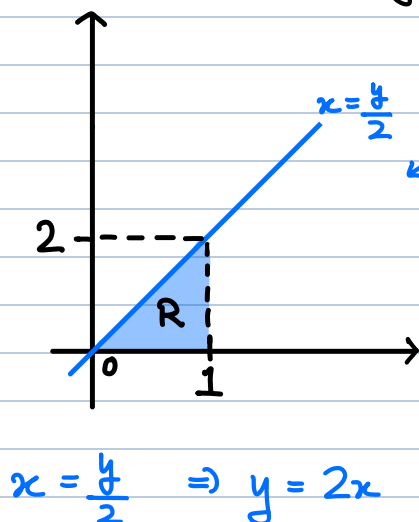
$$\begin{aligned}
&= \left[3x^2 - 9x + \frac{2x^3}{3} - x^2 - \frac{(-x+1)^4}{12} \right]_{-2}^0 \\
&\quad + \left[3x^2 - 9x - \frac{2x^3}{3} - x^2 + \frac{(x+1)^4}{12} \right]_0^2 \\
&= \left[\frac{-1}{12} - \frac{167}{12} \right] + \left[-\frac{103}{12} - \frac{1}{12} \right] = -\frac{68}{3} //
\end{aligned}$$

There might be times when you can't integrate the inner integral. At times like these, you'll need to reverse the order of integration.

(e.g.) Evaluate: $\int_0^2 \int_{y/2}^1 e^{x^2} dx dy$

Solⁿ Now $\int e^{x^2} dx$ is the error function. It can't be evaluated. So we need to reverse the order of integration: $\iint e^{x^2} dy dx$ ← But this will cause the limits to change as well

To find the new limits, we use the old limits to sketch the region R: $x = \frac{y}{2}$, $x = 1$, $y = 0$, $y = 2$



At $y = 2$, $x = \frac{y}{2} = 1$

∴ New limits from the

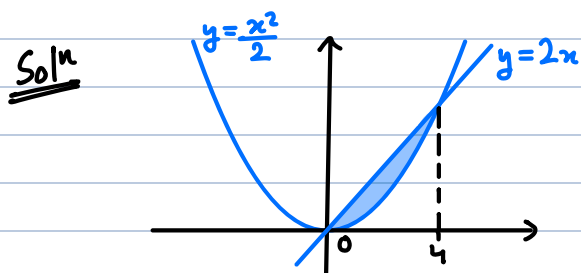
graph: $\int_0^1 \int_0^{2x} e^{x^2} dy dx$

$$= \int_0^1 [y e^{x^2}]_0^{2x} dx = \int_0^1 2x e^{x^2} dx$$

$$= [e^{x^2}]_0^1 = e - 1 //$$

(e.g.) Calculate the Area enclosed between the parabola $y = \frac{1}{2}x^2$ and $y = 2x$ using double integrals.

* Double integrals are usually used to find the volume.
But we can also use them to find the area R
(the region over which integration is done) by setting
the integrand = 1.



Points of intersection:

$$2x = \frac{x^2}{2} \Rightarrow x^2 - 4x = 0$$

$$\Rightarrow x(x-4) = 0 \Rightarrow x = 0 \mid \Rightarrow x = 4$$

$$\therefore \text{Area} = \int_0^4 \int_{\frac{x^2}{2}}^{2x} 1 \, dy \, dx$$

$$= \int_0^4 2x - \frac{x^2}{2} \, dx = \left[x^2 - \frac{x^3}{6} \right]_0^4 = \underline{\underline{\frac{16}{3}}}$$